Quantification: 699_HYP_HP_z25c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	8
interior lacuna count	8
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 8)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1207.43	1302	307.964	px^2
major_axis_length_px	lacuna shape	61.4549	60.9018	15.0293	px
minor_axis_length_px	lacuna shape	29.1411	30.4825	5.9332	px
aspect_ratio	lacuna shape	2.1753	2.0671	0.6798	unitless
eccentricity	lacuna shape	0.8561	0.8752	0.0787	unitless
solidity	lacuna shape	0.8154	0.7724	0.0713	unitless
roots_count	headline	9.125	5.5	13.984	count
ring_length_r30_px	headline	356.75	286.5	400.053	px
ring_length_r60_px	reported	684.5	581.5	675.909	px
owned_length_px	ownership-dependent	893.865	431.963	1427.56	px
edge_count	ownership-dependent	51.75	19	87.0808	count
mean_edge_length_px	ownership-dependent	18.7909	17.2822	10.2707	px
perimeter_px	normalised	341.071	200.673	384.104	px
ring_area_r30_px2	normalised	8502.88	6465	6785.76	px^2
ring_area_r60_px2	normalised	19994	15827.5	13064.7	px^2
in_frame_fraction_r30	normalised	0.9797	1	0.0574	unitless
in_frame_fraction_r60	normalised	0.9631	1	0.1043	unitless
ring_density_r30	normalised	0.0365607	0.0394448	0.01451	px^-1
ring_density_r60	normalised	0.0306072	0.0303755	0.0121612	px^-1
roots_per_100px_perimeter	normalised	2.2355	1.7662	1.641	per 100 px
ring_attached_length_r30_px	network v2	316	215.5	401.091	px
ring_attached_length_r60_px	network v2	544.5	397	679.096	px
ring_length_w_r30_px	network v2	400.763	322.85	451.538	px
ring_length_w_r60_px	network v2	765.265	644.123	760.299	px
sholl_crossings_r10	network v2	8.875	7	10.2739	count
sholl_crossings_r20	network v2	10.375	9.5	7.9631	count
sholl_crossings_r30	network v2	8.625	6.5	8.8146	count
loop_count	loops, report only	3.75	0	8.7627	count
loop_edge_share	loops, report only	0.0778	0	0.147	share
ring_width_mean_r30_px	width	6.9671	6.5684	1.0245	px

Per field
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measure	value	units
analysed_area_px2	896890	px^2
total_skeleton_length_px	14846	px
canalicular_length_density_per_px	0.0165528	px^-1
junction_count	279	count
junction_density_per_px2	0.00031107	px^-2
lacuna_count	8	count

lacunae_per_px2	8.92e-06	px^-2
skeleton_component_count	306	count
mean_component_length_px	48.5163	px
median_component_length_px	17	px
field_length_density_w_per_px	0.0184862	px^-1
field_density_without_flagged_per_px	0.012855	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	785	px
unattached_component_count	39	count
canalicular_length_density_attached_per_px	0.0165528	px^-1
field_length_density_with_unattached_per_px	0.017428	px^-1
excluded_area_px2	134576	px^2
analysed_field_area_px2	914000	px^2
loop_count	33	count
loop_edge_share	0.1263	share
edge_count	982	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.9968	px
width_median_px	7.2111	px
width_p10_px	4	px
width_p90_px	10	px
width_pixels_used	12316	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1197	51.8325	30.7893	1.6835	0.8045	0.9089	-1.2998	286.212	636.439	False
2	1418	83.5326	24.5778	3.3987	0.9557	0.7698	-1.1714	319.255	630.773	False
3	1476	60.9018	39.0949	1.5578	0.7668	0.7644	-0.4297	540.3	441.844	False
4	1461	75.7643	32.2675	2.348	0.9048	0.7334	-1.4732	556.736	495.522	False
5	1302	67.6703	25.7149	2.6316	0.925	0.8906	1.4126	532.052	690.266	False
6	950	46.9475	30.4825	1.5401	0.7605	0.7724	-1.05	698.178	753.763	False
7	648	43.5355	21.061	2.0671	0.8752	0.8686	-1.0803	997.846	882.89	False
8								274.548	207.774	False

Network per lacuna

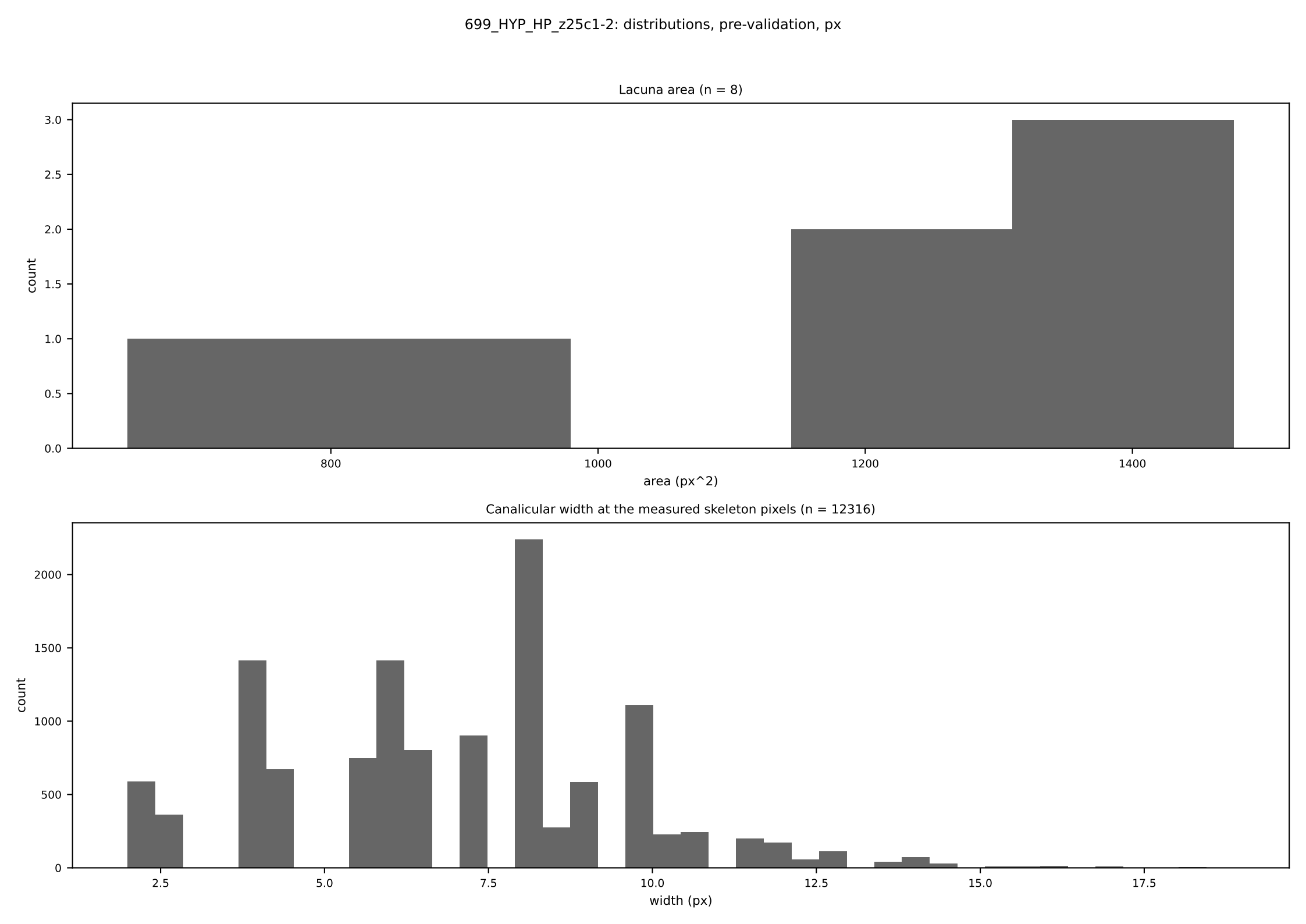
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	0	74	242	0	0	0	6.069
2	6	183	427	210.095	12	17.508	6.055
3	5	318	580	545.546	24	22.7311	8.304
4	6	309	629	1193.95	70	17.0564	7.4716
5	2	299	583	318.38	14	22.7414	6.8704
6	9	274	635	545.789	33	16.5391	6.1521
7	2	81	95	37.2132	1	37.2132	6.2663
8	43	1316	2285	4299.94	260	16.5382	8.5487

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	150.167	4208	10231	1	1	0.0175855	0.0236536	0
2	231.421	6473	15957	1	1	0.0282713	0.0267594	2.5927
3	282.806	6457	15698	1	1	0.0492489	0.0369474	1.768
4	340.049	6155	14117	1	1	0.0502031	0.0445562	1.7645
5	169.924	7529	20703	1	1	0.0397131	0.0281602	1.177
6	163.924	6994	19484	1	1	0.0391764	0.0325908	5.4904
7	116.569	5111	12557	0.8377	0.7049	0.0158482	0.0075655	1.7157
8	1273.71	25096	51205	1	1	0.0524386	0.0446245	3.376

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	24	77	82.1838	270.137	1	3	4
2	143	271	204.681	471.191	6	5	6
3	317	453	359.421	650.316	9	11	7
4	274	462	343.208	708.63	9	10	6
5	194	341	340.149	637.931	4	17	7
6	237	494	305.551	714.571	8	9	7
7	62	82	87.7696	103.598	1	2	2
8	1277	2176	1483.14	2565.75	33	26	30


699_HYP_HP_z25c1-2: distributions, pre-validation, px
Lacuna area (n = 8)
count
area (px^2)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
800
1000
1200
1400
Canalicular width at the measured skeleton pixels (n = 12316)
count
width (px)
0
500
1000
1500
2000
2.5
5.0
7.5
10.0
12.5
15.0
17.5